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8

ANALYTICAL METHODS FOR TRANSITION REGIME FLOWS

8.1 Classification of methods

A *transition regime flow*[†] is defined here as one in which the mean free path is neither very small nor very large in comparison with a typical linear dimension of the flow. The mathematical difficulties associated with the full Boltzmann equation generally preclude direct approaches that would lead to exact closed-form solutions. Moreover, for flows that involve chemical reactions and thermal radiation, there is difficulty even in the formulation of a Boltzmann equation that includes these effects. This has meant that a large number of indirect approaches have been investigated, and many of the resulting methods involve a degree of approximation that would not be accepted for less difficult equations. It is therefore necessary to distinguish between and to assess the results from a wide variety of competing methods. The answers to the following questions assist with the evaluation process:

(*i*) Does the method depend on arbitrary assumptions about the functional form of certain parameters, or even on arbitrary modifications of the equations?

(*ii*) Does the method lead to expressions for the dependent variables, or does it merely transform the equations into forms that are claimed to be more amenable to solution (either analytical or numerical) than the original equations?

(*iii*) Is the method capable of handling all values of all parameters, or is it restricted to limiting cases in which one of the parameters is either very large or very small?

(*iv*) What is the degree of complexity in terms of real gas effects, boundary conditions, and number of flowfield dimensions that can be handled by the method?

(*v*) Does the method require an initial approximation to the complete flowfield and depend on the convergence of a subsequent iteration process?

[†] This should not be confused with the 'transition' between laminar and turbulent flow and, to avoid this possibility, some authors refer instead to 'transitional' regime flows.

When any degree of approximation is involved, careful comparisons must be made with the results from other analytical and numerical methods, and with experiment. This is particularly important when the answer to question (*i*) is affirmative. General conclusions should not be based on comparisons for a single case, or even for a single problem. Also, comparative tests should, as far as possible, be based on the full range of flow quantities over the complete flowfield, rather than on the variation of a single integrated quantity with Knudsen number. The fact that solutions are available for the two limiting cases of collisionless and continuum flows means that superficially good results may be obtained from physically unreal methods that happen to provide a fortuitously good curve fit between these known limits. Particular scrutiny should be given to solutions that are based on approximations which introduce adjustable parameters.

With reference to the second question, it should be noted that some analytical methods provide 'solutions' that may require extensive numerical work in order to produce a result for a specific flow. In some cases, this numerical effort may exceed that which would be required for a direct numerical solution of the problem. These methods should then be regarded as providing alternative and perhaps less exact formulations rather than solutions.

Problems which involve large disturbances are invariably non-linear and there is then no analytical method that does not incorporate assumptions or approximations. The small-perturbation approach may be based on the smallness of the Knudsen number or its reciprocal, in order to obtain approximate solutions for the near continuum and near collisionless regimes, respectively. Some methods may also require that the speed ratio be small or large, or that the flow disturbance itself be small. In particular, if the disturbance is sufficiently small that the velocity distribution function is perturbed only slightly from the equilibrium or Maxwellian form, the small-perturbation approach leads to the linearized Boltzmann equation. The theory of this equation has been dealt with in detail by Cercignani (1969, 1988). The small-perturbation solutions provide important reference values, especially when they are obtained without resort to arbitrary assumptions about parameters or modifications to the equations. However, solutions are available only for a restricted set of problems with comparatively simple boundary conditions. These have limited application to problems that are encountered in practice.

A number of methods depend on essentially arbitrary assumptions about the form of the velocity distribution function. Most of these can be classed as *moment methods* and are discussed in §8.2. The other area of major approximation is to alter the form of the Boltzmann equation itself. It is indicative of the difficulties that have been posed by the Boltzmann equation that such a radical approach was adopted by a sufficient number of workers for it to have gained a measure of respectability that lasted for a considerable period. This *model equation* approach is discussed in §8.3.

8.2 Moment methods

This approach employs the moment equations that are obtained by first multiplying the Boltzmann equation by a molecular quantity Q, and then integrating it over velocity space. The moment equation for Q was derived as eqn (3.27) in the form

$$\frac{\partial}{\partial t}(n\bar{Q}) + \nabla.(n\overline{\boldsymbol{c}Q}) - n\boldsymbol{F}.\overline{\frac{\partial Q}{\partial \boldsymbol{c}}} = \Delta[Q]. \tag{8.1}$$

The macroscopic gas quantities were defined in §1.4 in terms of the averages of the microscopic molecular quantities Q. The substitution of the various values of Q into eqn (8.1) leads to a series of equations in the macroscopic quantities, as described in §3.3. However, the presence of $\overline{\boldsymbol{c}Q}$ in the second term means that, as Q progresses to successively higher orders of c, each equation involves a moment of still higher order. We therefore have an infinite number of equations and moments. Eqn (3.3) enables a typical moment to be written

$$\bar{Q} = \int_{-\infty}^{\infty} Q f \, \mathrm{d}\boldsymbol{c}$$

and the basis of the moment method is that f is assumed to conform to some expression that contains a finite set of macroscopic quantities or moments. The equation based on the moment that is of the highest order in c contains an even higher order moment, but this can now be written in terms of the lower-order moments, thus closing the series of equations and forming a determinate set.

The Chapman–Enskog solution of the Boltzmann equation has already been discussed in §3.5. It is the second-order solution based on a series expansion of the velocity distribution function f. This series may be written schematically as

$$f = f_0\{1 + a_1(Kn) + a_2(Kn)^2 + \dots\}, \tag{8.2}$$

where the coefficients a_n are functions of ρ, $\boldsymbol{c}_0$, and T only. The Knudsen number is defined here as the ratio of the scale length of the flow gradients to the mean free path. The first-order† solution is the local equilibrium or Maxwellian distribution function f_0, with the gas being fully described by ρ, $\boldsymbol{c}_0$, and T. The viscous stress tensor $\boldsymbol{\tau}$ and the heat flux vector $\boldsymbol{q}$ vanish in an equilibrium gas and the conservation equations (3.33)–(3.35) reduce to the Euler equations of inviscid fluid flow. The conservation equations are the five moment equations formed by setting Q equal to m, $m\boldsymbol{c}$, and $\frac{1}{2}mc^2$ and, since these quantities are conserved in collisions, the collision term $\Delta[Q]$ vanishes.

† As far as the powers of (Kn) are concerned, this is the zero-order solution. However, in §3.5, the 'order' of the solution was based on the number of terms in the series and this convention is continued here.

The second-order Chapman–Enskog solution leads to the velocity distribution function of eqn (3.80) which also involves only the moments ρ, $\boldsymbol{c}_0$, and T. This distribution function enables τ and $\boldsymbol{q}$ to be written as products of the coefficients μ and K with the velocity and temperature gradients, respectively, thus reducing the conservation equations to the Navier–Stokes equations of continuum gas dynamics.

The above considerations show that, from the kinetic theory point of view, both the Euler and Navier–Stokes equations may be regarded as 'five moment' solutions of the Boltzmann equation, the former being valid for the $(Kn) \to 0$ limit and the latter for $(Kn) << 1$. In the context of continuum gas dynamics, the Euler equations describe reversible adiabatic (isentropic) flows, while the Navier–Stokes equations provide the standard description of viscous flows. This means that the first- and second-order Chapman–Enskog solutions merely reconcile the molecular and continuum approaches for small Knudsen number flows. It is obvious that the expansion for f in eqn (8.2) must break down as the Knudsen number approaches unity, but there remains the possibility that the use of more than two terms may extend the range of applicability of the continuum approach to higher Knudsen numbers. The third-order solution of Eqn (8.2) leads to the Burnett equations. These were discussed in §3.5 and it was pointed out that there is evidence that they not only extend the range of validity of the Navier–Stokes equations, but can provide a more accurate description at all Knudsen numbers. They are, however, extremely complex and closed form solutions are not available even for the simple flows that are readily solved at the Navier–Stokes level of approximation. Moreover, at the higher Knudsen numbers, the numerical effort in solving the Burnett equations is far greater than that involved in obtaining accurate solutions from the direct simulation methods that are described in the later chapters.

Grad (1949) put forward an alternative expansion for f as a series of Hermite tensor polynomials with the first-order solution again being the local Maxwellian. The third-order solution leads to an expression for f as the local Maxwellian multiplied by an expression involving τ and $\boldsymbol{q}$ as well as ρ, $\boldsymbol{c}_0$, and T. This means that the total number of moment equations that are required for a determinate set is equal to the total number of dependent variables in the conservation equations. This number is readily seen to be thirteen, thus leading to Grad's Thirteen Moment equations. While these represented a logical and important development in kinetic theory, the results from their application to specific problems in the transition regime have proved to be disappointing (Schaaf and Chambre 1961).

The failure of the systematic expansion methods to produce equations that significantly extend the upper Knudsen number limits of the Navier–Stokes equations indicates that more radical steps must be taken if workable approximations are to be found for the distribution function in the transition regime. For one-dimensional steady flow problems with two-point boundary conditions, the procedure has been to choose f as a combination of the two Maxwellian distributions that apply at the boundaries.

As the first example, consider the 'four moment solution' of Liu and Lees (1961) for the steady one-dimensional heat flux between infinite flat plates. The lower plate in the plane $y = 0$ is at the temperature T_L and the upper plate in the plane $y = h$ is at the temperature T_U. The assumption that is made about the form of the distribution function is that it is a combination of two half-range Maxwellians, each with a different temperature. This may be expressed as

$$nf = N_1 f_1 + N_2 f_2, \tag{8.3}$$

where f_1 and f_2 are the distribution functions of the molecules moving in the positive and negative y-directions, respectively. The temperatures T_1 and T_2 of these Maxwellian distributions as well as the parameters N_1 and N_2 are functions of y.

The average of any molecular quantity Q is given by eqns (8.3), (3.3), and (4.1) as

$$\begin{aligned}\bar{Q} = (1/n)\{ & N_1 \beta_1^3 \pi^{-3/2} \int_{-\infty}^{\infty}\int_{0}^{\infty}\int_{-\infty}^{\infty} Q \exp(-\beta_1^2 c'^2)\, \mathrm{d}u\, \mathrm{d}v\, \mathrm{d}w \\ & + N_2 \beta_2^3 \pi^{-3/2} \int_{-\infty}^{\infty}\int_{-\infty}^{0}\int_{-\infty}^{\infty} Q \exp(-\beta_2^2 c'^2)\, \mathrm{d}u\, \mathrm{d}v\, \mathrm{d}w \}.\end{aligned} \tag{8.4}$$

The boundary conditions are such that the stream velocity must be zero and $c' = c$. Setting Q equal to m, v, c^2, and $\frac{1}{2}mvc^2$ yields the following expressions for the important macroscopic quantities in terms of the four unknown quantities N_1, N_2, β_1, and β_2:

$$\rho = nm = \tfrac{1}{2}(N_1 + N_2)m, \tag{8.5}$$

$$v_0 = \bar{v} = (N_1/\beta_1 - N_2/\beta_2)/(2n\pi^{1/2}), \tag{8.6}$$

$$T = \overline{c^2}/(3R) = (N_1/\beta_1^2 + N_2/\beta_2^2)/(4nR), \tag{8.7}$$

and

$$q_y = \tfrac{1}{2}nm\overline{vc^2} = m(N_1/\beta_1^3 - N_2/\beta_2^3)/(2\pi^{1/2}). \tag{8.8}$$

The moment equation for this problem is, from eqn (3.27)

$$\frac{\mathrm{d}}{\mathrm{d}y}\left(n\int_{-\infty}^{\infty} Qvf\, \mathrm{d}\boldsymbol{c}\right) = \Delta[Q]. \tag{8.9}$$

The conserved quantities m, mv, and $\frac{1}{2}mc^2$ provide obvious choices for Q in three of the four moment equations that are required. The collision integral vanishes for these equations and eqn (8.9) becomes

$$n\int_{-\infty}^{\infty} Qvf\, \mathrm{d}\boldsymbol{c} = \text{const}. \tag{8.10}$$

Setting $Q = m$, we obtain

$$N_1/\beta_1 - N_2/\beta_2 = 0. \tag{8.11}$$

The constant is zero in eqn (8.11) because the left-hand side may be directly related through eqn (8.6) to the stream velocity normal to the plates, and the boundary conditions require this to vanish at the plates. For $Q = v$, eqn (8.10) yields

$$N/\beta_1^2 + N_2/\beta_2^2 = \text{const}$$

and, using eqn (8.7),

$$N_1/\beta_1^2 + N_2/\beta_2^2 = 4p/m \tag{8.12}$$

with the pressure $p = nmRT$ a constant for this flow. Similarly, with $Q = \frac{1}{2}mc^2$, and using eqn (8.8)

$$N_1/\beta_1^3 - N_2/\beta_2^3 = 2\pi^{\frac{1}{2}} q_y/m. \tag{8.13}$$

Because the problem essentially involves heat transfer in the y-direction, Lees and Liu chose $Q = \frac{1}{2}mvc^2$ for the fourth moment. Eqn (8.9) then becomes

$$\frac{5m}{16}\frac{\mathrm{d}}{\mathrm{d}y}\left(\frac{N_1}{\beta_1^4} + \frac{N_2}{\beta_2^4}\right) = \Delta[\tfrac{1}{2}mvc^2].$$

The collision integral for $\frac{1}{2}mvc^2$ may be evaluated in a similar manner to that for u^2 in §3.3 and, for the special case of Maxwell molecules the result is (Vincenti and Kruger 1965, p. 364)

$$\Delta[\tfrac{1}{2}mvc^2] = -\pi A_2(5)(2\kappa/m)nq_y.$$

The constant $A_2(5)$ and the parameter κ may be eliminated by eqns (3.56) and (3.64) for the coefficient of heat conduction in a gas of Maxwell molecules. The fourth moment equation then becomes

$$\frac{\mathrm{d}}{\mathrm{d}y}\left(\frac{N_1}{\beta_1^4} + \frac{N_2}{\beta_2^4}\right) = -\frac{8Rpq_y}{mK}. \tag{8.14}$$

Now, from eqns (8.5), (8.7), and (8.11)

$$T = p/(nmR) = (2R\beta_1\beta_2)^{-1} = (T_1T_2)^{\frac{1}{2}} \tag{8.15}$$

which, with eqns (8.8), (8.11), and (8.12), gives

$$\frac{1}{\beta_1} - \frac{1}{\beta_2} = \frac{\pi^{\frac{1}{2}} q_y}{2p} \tag{8.16}$$

and

$$\frac{N_1}{\beta_1^4} + \frac{N_2}{\beta_2^4} = \frac{4p}{m}\left(\frac{\pi q_y^2}{4p^2} + 2RT\right).$$

Eqn (8.14) can therefore be written

$$q_y = -K\frac{dT}{dy} \tag{8.17}$$

which is, of course, the standard continuum result. The four moment solution departs from the continuum solution only through the boundary conditions which are

$$T_1 = T_L \quad \text{at } y = 0$$

and

$$T_2 = T_U \quad \text{at } y = h. \tag{8.18}$$

Eqns (8.15) and (8.16) then give

$$T_0 = T_L - \tfrac{1}{2}(\tfrac{1}{2}\pi T_L/R)^{\frac{1}{2}} q_y/p$$

and

$$T_h = T_U + \tfrac{1}{2}(\tfrac{1}{2}\pi T_U/R)^{\frac{1}{2}} q_y/p \tag{8.19}$$

for the temperature T_0 at $y = 0$ and T_h at $y = h$.

For a gas of Maxwell molecules, $K = CT$ and the solution of eqn (8.17) that was given in eqn (7.25) becomes

$$q_y = -\tfrac{1}{2}C(T_h^2 - T_0^2)/h. \tag{8.20}$$

Eqns (8.19) and (8.20) form a determinate set for the solution of q_y, T_0, and T_h. The result for the heat flux is best expressed as the ratio of q_y to the free-molecule flux q_f as a function of the overall Knudsen number based on the average value of the mean free path and the plate spacing. Eqns (4.52) and (3.56) enable the mean free path to be written

$$\lambda = \frac{8}{15}\left(\frac{1}{2\pi RT}\right)^{\frac{1}{2}}\frac{K}{R\rho} = \frac{8}{15}\left(\frac{1}{2\pi RT}\right)^{\frac{1}{2}}\frac{CT}{R\rho}. \tag{8.21}$$

Then, noting that the pressure is constant in this flow, the Knudsen number based on the average mean free path in the gas between the plates is

$$(Kn) = \frac{\bar{\lambda}}{h} = \frac{8C\overline{T^{3/2}}}{15h(2\pi R)^{\frac{1}{2}}p}. \tag{8.22}$$

The equations may be solved numerically and the results for $T_U/T_L = 4$ are shown in Fig. 8.1. The transition from continuum to free-molecule flow is, in this case, centred on an overall Knudsen number of 0.2 and spans more than a thousandfold change in this number. There is a 1% departure from the continuum solution at $(Kn) = 0.0018$, and the heat flux reaches 99% of the free-molecule value at $(Kn) = 140$. The empirical 'bridging formula' approach to transition regime flows generally constructs a smooth curve between the continuum solution at a Knudsen number of 0.1 and the free-molecule solution at a Knudsen number of 10. This approach is not supported by the four moment solution for this problem.

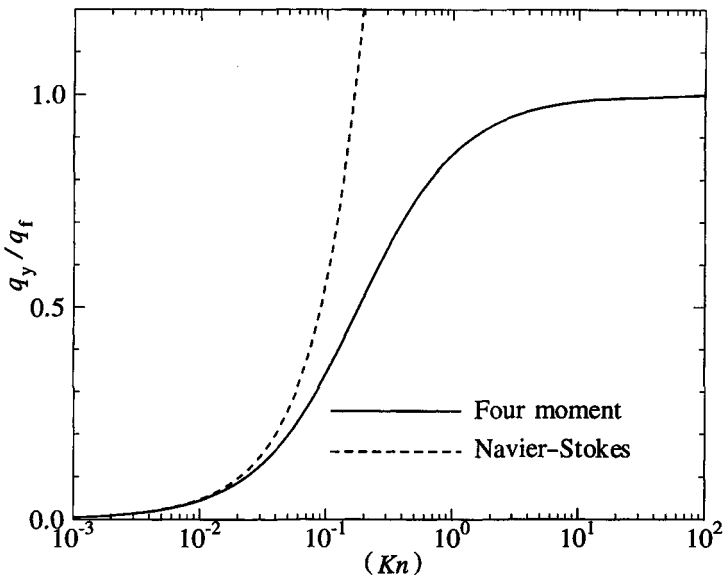


Fig. 8.1 The four moment solution for the one-dimensional heat transfer between parallel plates with a temperature ratio of four.

The temperature jump or 'slip' at a surface at temperature T_p can be written in terms of the *local* Knudsen number (Kn') which is defined as the ratio of the local mean free path to the scale length of the temperature gradient. The resulting temperature jump is

$$T - T_p = \pm \frac{15\pi}{16} (T_p T)^{1/2} \frac{\lambda}{T} \frac{dT}{dy} = \pm \frac{15\pi}{16} (T_p T)^{1/2} (Kn'), \qquad (8.23)$$

with the positive and negative signs applying at the lower and upper plates, respectively. The difference between the temperatures of the upward and downward moving molecules can be written

$$T_1 - T_2 = -\frac{15\pi}{16} \left\{ (T_1 T)^{1/2} + (T_2 T)^{1/2} \right\} \frac{\lambda}{T} \frac{dT}{dy}, \qquad (8.24)$$

and is consistent with the result for the temperature jump.

Eqn (8.23) shows that the temperature jump tends to zero with the Knudsen number, and the heat transfer tends to the continuum value. However, the assumed distribution function of eqn (8.3) does not tend to the Chapman–Enskog distribution function. This can be illustrated through the calculation of the difference between the separate temperatures of the upward and downward moving molecules in the Chapman–Enskog gas. The temperature T^+ of the upward moving molecules in this

gas is obtained by setting the stream velocity in eqn (4.57) to zero, setting this expression into eqns (3.3) and (1.29), and performing the integration over v from 0 to ∞. The temperature of the downward moving molecules is obtained similarly, and the temperature difference between the upward and downward moving molecules is

$$\frac{T^+ - T^-}{T} = -\frac{5(\alpha+1)(\alpha+2)}{\alpha(5-2\omega)(7-2\omega)}\frac{\lambda}{T}\frac{dT}{dy}. \tag{8.25}$$

The Maxwell gas case is given by $\omega = 1$, and a comparison of eqns (8.24) and (8.25) shows that they then differ by more than a factor of two. It is, therefore, incorrect to say that the four moment solution is in agreement with the Navier–Stokes solution at small values of the overall Knudsen number. The Chapman–Enskog result would obviously be more accurate well away from the surface, but the moment solution employs the correct distribution for molecules that are diffusely reflected from the wall. However, it can be safely concluded that the temperature jump ratio is of the order of the local Knudsen number. If the u_0 stream velocity component varies in the y-direction, there is also a difference in the stream velocity of the upward and downward moving molecules. This is obtained in a similar manner to eqn (8.25) as

$$\frac{u^+ - u^-}{u} = -\frac{5(\alpha+1)(\alpha+2)}{2\alpha(5-2\omega)(7-2\omega)}\frac{\lambda}{u}\frac{du}{dy}. \tag{8.26}$$

This shows that the *velocity slip* ratio at the surface is of the order of the local Knudsen number based on the scale length of the gradient in the stream velocity. Continuum flow solutions that are based on Navier–Stokes equations are improved if the usual 'no-slip' boundary conditions are modified to allow for the slips in velocity and temperature that occur at solid surfaces. The Chapman–Enskog distribution adjusts to the surface distribution in a region, called the *Knudsen layer*, that has a thickness of the order of the mean free path.

Probably the best-known moment method is that of Mott-Smith (1951) for shock wave structure. This is also a one-dimensional flow and it is assumed that the distribution function within the wave may be represented as a linear combination of the distribution functions that apply in the uniform upstream and downstream flows. This is a bimodal distribution and can be written

$$nf = N_1 f_1 + N_2 f_2, \tag{8.27}$$

where the subscripts $_1$ and $_2$ denote the uniform upstream and downstream conditions, respectively. This equation is identical to eqn (8.3) for the four moment distribution function but, in that case, f_1 and f_2 were variables rather than constants, and the distribution could not be described as bimodal. The x-axis may be taken normal to the wave and eqn (4.1) gives

$$f_1 = (\beta_1^2/\pi^{3/2}) \exp[-\beta_1^2\{(u-u_{01})^2 + v^2 + w^2\}]$$

and (8.28)

$$f_2 = (\beta_2^2/\pi^{3/2}) \exp[-\beta_2^2\{(u-u_{01})^2 + v^2 + w^2\}] .$$

The weighting factors N_1 and N_2 must be such that $N_1 = n_1$ and $N_2 = 0$ upstream of the wave, while $N_1 = 0$ and $N_2 = n_2$ downstream of the wave. Eqn (8.27) may be integrated over velocity space and the normalization condition of eqn (3.2) then requires

$$n = N_1 + N_2 . \tag{8.29}$$

The application of the conservation equations between the upstream and downstream conditions leads to the Rankine–Hugoniot 'jump' equations for n_2, u_{02}, and β_2 as functions of n_1, u_{01}, and β_1. The evaluation of all the flow quantities within the wave as the appropriate moments of the bimodal distribution function of eqn (8.27) requires a solution for n, N_1, and N_2 as functions of position within the wave. One relation between these quantities has already been provided by eqn (8.29). A second is obtained by applying the mass conservation equations between the upstream state and a point within the wave. Eqn (3.30) gives

$$n_1 u_{01} = n u_0 \tag{8.30}$$

and, evaluating $u_0 = \int_{-\infty}^{\infty} uf \, d\boldsymbol{c}$ for the distribution defined by eqn (8.27),

$$n_1 u_{01} = N_1 u_{01} + N_2 u_{02} . \tag{8.31}$$

The third relation is supplied by an additional moment equation. The choice of the quantity Q in this equation is arbitrary; Mott-Smith used both u^2 and u^3. With $Q = u^2$, eqn (3.27) for a steady one-dimensional flow is

$$\frac{d}{dx}(n\overline{u^3}) = \Delta[u^2]$$

and, using eqn (3.66) for the collision integral in a gas of either Maxwell molecules or VHS molecules with $\upsilon = ½$,

$$\frac{d}{dx}(n\overline{u^3}) = \frac{p}{m}\frac{\tau_{xx}}{\mu} . \tag{8.32}$$

Eqns (3.64) and (3.65) show that the viscosity coefficient for these gas models can be written CT where C is a constant. Eqn (4.52) for the mean free path† enables C to be written in terms of the upstream conditions, i.e.

$$C = \frac{\rho_1 \lambda_1}{2}\left(\frac{2\pi k}{m T_1}\right)^{½} .$$

This may be substituted into eqn (8.32), to give

† This book employs the mean free path of eqn (4.52) that takes into account the change in collision cross-section with relative velocity. This causes the results to differ from earlier treatments that employ the hard sphere value by the factor $(5-2\omega)(7-2\omega)/24$, or 5/8 for Maxwell model that is employed here.

$$\frac{\mathrm{d}}{\mathrm{d}x}(n\overline{u^3}) = \frac{(2RT_1/\pi)^{1/2}}{n_1\lambda_1}\frac{n}{m}\tau_{xx}. \tag{8.33}$$

Now,

$$n\overline{u^3} = N_1\int_{-\infty}^{\infty} u^3 f_1\,\mathrm{d}\boldsymbol{c} + N_2\int_{-\infty}^{\infty} u^3 f_2\,\mathrm{d}\boldsymbol{c}$$

and

$$\begin{aligned}\int_{-\infty}^{\infty} u^3 f_1\,\mathrm{d}\boldsymbol{c} &= \frac{\beta_1^3}{\pi^{3/2}}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\{u_{01}+(u-u_{01})\}^3\exp[-\beta_1^2\{(u-u_{01})^2\\ &\quad + v^2+w^2\}]\,\mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w\\ &= u_{01}(u_{01}^2+3RT).\end{aligned}$$

A similar expression may be obtained for the second term and, using eqn (8.30)

$$n\overline{u^3} = N_1u_{01}(u_{01}^2+3RT_1-u_{02}^2-3RT_2)+n_1u_{01}(u_{02}^2+3RT_2).$$

Conservation of energy between the upstream and downstream states requires

$$u_{01}^2+5RT_1 = u_{02}^2+5RT_2,$$

so that

$$\frac{\mathrm{d}}{\mathrm{d}x}(n\overline{u^3}) = \frac{2}{5}u_{01}(u_{01}^2-u_{02}^2)\frac{\mathrm{d}N_1}{\mathrm{d}x}. \tag{8.34}$$

Also, from eqn (1.26) and noting that τ_{xx} is symmetrical about the x-axis,

$$(n/m)\tau_{xx} = n^2(\overline{c'^2}/3-\overline{u'^2}) = 2n^2(\overline{v'^2}-\overline{u'^2})/3.$$

Then, similarly to the derivation of eqn (1.29a) from (1.29),

$$\overline{u'^2} = \overline{u^2}-u_0^2$$

and the mean value of u^2 may be evaluated in a similar manner to the above treatment of the mean value of u^3. This gives,

$$(n/m)\,\tau_{xx} = 2n(nu_0^2-N_1u_{01}^2-N_2u_{02}^2)/3$$

and, using eqns (8.29), (8.30), and (8.31),

$$\begin{aligned}(n/m)\,\tau_{xx} &= 2\{(N_1u_{01}+N_2u_{01})^2-N_1(N_1+N_2)u_{01}^2-N_2(N_1+n_2)u_{02}^2\}/3\\ &= -2N_1N_2(u_{01}-u_{02})^2/3.\end{aligned}$$

N_2 may again be eliminated through eqn (8.31) to give

$$\frac{n}{m}\tau_{xx} = -\frac{2}{3}N_1(n_1-N_1)\frac{u_{01}}{u_{02}}(u_{01}-u_{02})^2. \tag{8.35}$$

The substitution of eqns (8.34) and (8.35) into (8.33) leads to the following differential equation for N_1/n_1 as a function of x/λ_1

$$\frac{\mathrm{d}}{\mathrm{d}(x/\lambda_1)}\left(\frac{N_1}{n_1}\right) = -\alpha\frac{N_1}{n_1}\left(1-\frac{N_1}{n_1}\right), \tag{8.36}$$

where

$$\alpha = \frac{5}{3\pi^{1/2}}\frac{(2RT_1)^{1/2}}{u_{02}}\frac{u_{01}-u_{02}}{u_{01}+u_{02}}. \tag{8.37}$$

The solution of eqn (8.36) is

$$N_1 = \frac{n_1}{1+\exp\{\alpha(x/\lambda_1)\}} \tag{8.38}$$

and the substitution of this and eqn (8.31) into eqn (8.27) provides an expression for the distribution function within the wave as a function of x/λ_1. The density profile follows directly from eqn (8.28) as

$$\frac{n-n_2}{n_1-n_2} = \frac{N_1}{n_1} = \frac{1}{1+\exp\{\alpha(x/\lambda_1)\}}. \tag{8.39}$$

The profiles of the other macroscopic flow quantities may be obtained as the appropriate moment of the bimodal distribution function.

The Mott-Smith solution has been found to be inadequate for the weaker (shock Mach numbers less than about 1.6) shock waves that are adequately described by the Navier–Stokes equations. However, it will be shown in §12.11 that the bimodal distribution function is qualitatively correct for very strong shock waves, and the Mott-Smith method is reasonably successful in predicting the structure of such waves. The shock thickness L is well predicted when the molecular model is chosen to match the coefficient of viscosity, but the profiles of real waves are not symmetrical as is predicted by the theory. It must also be remembered that the good result for the thickness of strong waves is based on the choice of u^2 as the quantity Q in the fourth moment equation. This is the usual choice, but there is no rigorous justification for it, and Rode and Tanenbaum (1967) and Sather (1973) have shown that the result is highly sensitive to it. It may be noted that, concurrently with the experimental justification of the result with $Q = u^2$, an argument was presented (Desphande and Narasimha 1969) to the effect that the u^3 result should be superior to the u^2 result.

The method was extended to the other inverse power law molecular models by Muckenfuss (1962). This made it possible to match the real gas temperature exponent, as well as the actual value, of the coefficient of viscosity and improved the agreement with measured values. The shock thickness was found to be related to the exponent η in the inverse power law model and the shock Mach number $(Ma)_S$ by

$$L/\lambda_1 \propto (Ma)_S^{4/(\eta-1)}. \tag{8.40}$$

The justification for the VHS molecular model is that the observable effects of molecular model can be fully related to the change in the collision cross-section with the relative speed. The significant collisions in the shock structure problem are those that occur with relative speeds that are characteristic of the downstream temperature T_2. The temperature ratio across the shock is proportional to $(Ma)_S^2$ so that, from eqns (3.61) and (3.68),

$$\frac{(\sigma_T)_2}{(\sigma_T)_1} \propto \left(\frac{T_2}{T_1}\right)^{-\upsilon} \propto (Ma)_S^{-4/(\eta-1)}. \tag{8.41}$$

The increase in shock thickness with shock Mach number therefore correlates exactly with the reduction in the relevant collision cross-section.

The moment method has been applied successfully to other steady one-dimensional problems such as condensation and evaporation (Ytrehus 1994). While the results from some applications of the method are in remarkably good agreement with experiment, the solutions are not unique because the method is based on several arbitrary choices. One of these is associated with the form of the assumed distribution function, and the other with the choice of the non-conserved quantities for the additional moment equations. The best choices for one set of parameters is not necessarily the best for another set, even for the same problem. Also, the use of a more complicated distribution function, with a consequent increase in the number of moments, does not necessarily lead to more consistent results. The moment methods have not been applied to two-dimensional or axially symmetric problems such as the steady transition regime flow past a sphere. The major problem is that the form of the distribution function varies greatly from one location to another in such a flow. The difficulty in formulating a suitable expression for the distribution function is further complicated by the fact that, for these flows, it is three-dimensional rather than axially symmetric in velocity space. Finally, even if it were possible to find a workable approximation to the distribution function, the resulting equations would be more difficult to solve than the Navier-Stokes equations for the corresponding viscous continuum flow.

8.3 Model equations

The model equation approach involves an approximation to the form of the Boltzmann equation itself. It is the collision term on the right hand side of the Boltzmann equation that poses the greatest mathematical difficulties, and it is this term that is modified. The best known model equation is called the BGK equation after Bhatnagar, Gross, and Krook (1954). This may be written

$$\frac{\partial}{\partial t}(nf) + \boldsymbol{c}.\frac{\partial}{\partial \boldsymbol{r}}(nf) + \boldsymbol{F}.\frac{\partial}{\partial \boldsymbol{c}}(nf) = n\nu(f_0 - f). \tag{8.42}$$

While the parameter ν is generally regarded as a collision frequency, it has a restricted functional dependence in that it is proportional to density and

may depend on temperature, but is assumed to be independent of the molecular velocity $\boldsymbol{c}$. The appearance of the local Maxwellian f_0 means that the equation remains a non-linear integro-differential equation. This is because f_0 is a function of the stream velocity $\boldsymbol{c}_0$ and the temperature T, and these must be obtained as integrals over f. The integration over both f_0 and f must result in identical values of $\boldsymbol{c}_0$ and T so that the collision term vanishes when moments are taken over the conserved quantities. This condition ensures that the BGK equation is consistent with the conservation equations.

The BGK equation for a stationary homogeneous gas without an external force field follows from eqn (8.42) as

$$\frac{\partial f}{\partial t} = \nu (f_0 - f)$$

and clearly has the correct solution $f = f_0$ at equilibrium. The Chapman–Enskog method may be applied to the equation (Vincenti and Kruger 1965, p. 379). This converts the conservation equations to the form of the Navier–Stokes equations, although the numerical constants in the transport properties do not correspond to any of the recognised molecular models. The BGK equation obviously provides the correct collisionless or free-molecule solution, since the form of the collision term is then immaterial. An exact solution would therefore span the complete range of Knudsen number and would be at or close to the correct limits at the extremes of this range. The approximate collision term would, however, lead to an indeterminate error in the transition regime.

A comparison of eqns (8.42) and (3.20) shows that the depletion component of the collision term in the Boltzmann equation

$$-\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 f f_1 c_r \sigma \, d\Omega \, dc_1$$

is replaced in the BGK equation by

$$-n\nu f.$$

The correct velocity dependent collision frequency $\nu(\boldsymbol{c})$ follows directly from the Boltzmann term as

$$\nu(\boldsymbol{c}) = n\int_{-\infty}^{\infty}\int_{0}^{4\pi} f_1 c_1 \sigma \, d\Omega \, dc_1 . \tag{8.43}$$

This is independent of molecular velocity only for the physically unreal Maxwell molecules, or for VHS molecules with $\omega = 1$. The use of a velocity independent ν in the BGK equation is generally an unjustified approximation, but at least the form of the depletion term is correct. The same cannot be said for the component of the collision term that represents the replenishing collisions. This is written

$$n\nu f_0$$

and assumes that, irrespective of the form of f, the number of molecules scattered by collisions into class $\boldsymbol{c}$ is equal to the number that would be scattered out of this class in an equilibrium gas having the assumed velocity independent collision frequency.

Liepmann et al. (1962) applied the BGK equation to the shock structure problem. For one-dimensional steady flow, eqn (8.42) becomes

$$u\frac{\mathrm{d}}{\mathrm{d}x}(nf) = n\nu(f_0 - f). \tag{8.44}$$

This may be integrated to give the following integral equation

$$nf = \int_{\mp\infty}^{x} \frac{n\nu f_0(x')}{u} \exp\left(-\int_{x'}^{x} \frac{\nu\,\mathrm{d}x''}{u}\right)\mathrm{d}x', \tag{8.45}$$

where x' and x'' are dummy variables and the negative and positive signs on the lower limit apply for positive and negative u, respectively. Eqn (8.45) was solved numerically by Liepmann et al. using an iterative scheme that required an initial guess for f_0. The results are in good agreement with the Navier–Stokes solution, and therefore with experiment, for shock Mach numbers significantly less than two. For very strong shock waves, the method leads to highly asymmetric velocity and density profiles in which the upstream region is far more extended than in the measured profiles. The probable reason for the failure of the BGK model in the upstream region of a very strong shock wave lies in the use of a collision frequency that is independent of f. This collision frequency is related to T through the BGK result for the coefficient of viscosity but, since this relationship is based on near-equilibrium theory, it is not adequate in the region of extreme non-equilibrium near the leading edge of the wave.

The assumptions inherent in the BGK equation are least likely to have deleterious effects in problems that involve small perturbations and have their boundary conditions prescribed in terms of equilibrium distributions. The distribution function then departs only very slightly from the Maxwellian form, irrespective of the value of the Knudsen number. This leads to linearized forms of the Boltzmann and BGK equations and, for these, the comparative mathematical simplicity of the model equations is particularly significant. In addition, the temperature variations and therefore the changes in the collision cross-sections are small, and the accuracy of the molecular model is not of critical importance. Most of the results in Cercignani (1969 and 1988) are for the BGK equation rather than for the full Boltzmann equation. There are a number of problems of practical importance that involve small perturbations. As noted by Cercignani (1991), these generally involve low Mach numbers and include a number of environmental problems. However, for non-linear problems that involve large perturbations and complex physical effects, analytical solutions are out of the question and recourse must be had to numerical methods.

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